

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME:
Networks

CSA0716 COURSE CODE: Computer

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I _____SHARVESH S(192525458)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SHARVESH S

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
3. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
4. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.
5. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.
7. An FTP application transfers **3 GB** of data over a network with an effective

throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions

Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

ANSWERS

1:ANS

Given:

File size = 150 MB

Bandwidth = 50 Mbps

Segment size = 1500 bytes

Data Link overhead = 40 bytes/frame

1. Number of Segments

Answer: 100,000 segments

2. Number of Frames

Each segment is placed into one frame.

Answer: 100,000 frames

3. Total Data Link Layer Overhead

Overhead per frame = 40 bytes

Answer: 4 MB overhead

4. Total Transmission Time

Total data transmitted:

Convert to bits:

Transmission time:

Answer: 24.64 seconds

5. Reliable Communication

Transport Layer: Divides the file into segments, checks for errors, retransmits lost segments, and ensures data arrives in the correct order.

Data Link Layer: Adds headers and trailers, detects frame errors using error checking (CRC), and ensures reliable delivery between directly connected devices.

2.ANS:

Given:

Sliding window size = 6 frames

Data per frame = 1024 bytes

Total frames = 48 frames

1. Number of Transmission Windows

Answer: 8 transmission windows

2. Total Retransmissions

One frame is lost in every window.

Answer: 8 retransmissions

3. How Flow Control Improves Communication Efficiency

Flow control prevents the sender from sending data faster than the receiver can process.

It avoids buffer overflow at the receiver.

It reduces data loss and unnecessary retransmissions.

It ensures smooth and efficient communication.

3.ANS:

Given:

Sliding window size = 6 frames

Each frame = 1024 bytes

Total frames = 48

1. Number of Transmission Windows

windows

2. Total Retransmissions

One frame is lost in each window. : transmissions

4.ANS:

Given:

Sliding window size = 6 frames

Total frames = 48

1. Number of Transmission Windows

$48 \div 6 = 8$ windows

Answer: 8 windows

2. Total Retransmissions

One frame is lost in each window.

$$8 \times 1 = 8 \text{ retransmissions}$$

Answer: 8 retransmissions

5.ANS:

Given:

Video file = 600 MB

Checkpoint every 60 MB

Failure after 390 MB

1. Number of Checkpoints Before Failure

Answer: 6 checkpoints

2. Data to be Retransmitted

Last checkpoint = 360 MB

Failure = 390 MB

Answer: 30 MB

3. Synchronization

Synchronization creates checkpoints during data transfer.

If a failure occurs, transmission restarts from the last checkpoint instead of the beginning.

This saves time and reduces data retransmission.

6.ANS:

Given:

Original file = 900 MB

Compression = 40%

Encryption overhead = 5%

1. Compressed File Size

Answer: 540 MB

2. Encrypted File Size

Answer: 567 MB

3. Total Percentage Reduction

Answer: 37% reduction

4. Presentation Layer

It compresses data to reduce file size.

It encrypts data to keep it secure.

This improves transmission efficiency and security.

7.ANS:

Given:

Data = 3 GB = 3000 MB

Request size = 3 MB

Throughput = 120 Mbps

1. Number of Requests

Answer: 1000 requests

2. Transmission Time

Total data = $3000 \times 3 = 9000$ MB

Answer: 200 seconds

3. Application Layer

It provides services like FTP for file transfer.

It ensures reliable communication between users and applications.

It manages file transfer requests efficiently.

8.ANS:

Given:

Application Layer = 4 ms

Transport Layer = 3 ms

Network Layer = 5 ms

Data Link Layer = 2 ms

Physical Layer = 1 ms

Propagation Delay = 18 ms

Transmission Delay = 12 ms

1. Total End-to-End Delay

Answer: 45 ms

2. Layer Contribution

Application Layer: Provides network services.

Transport Layer: Ensures reliable data transfer.

Network Layer: Routes packets to the destination.

Data Link Layer: Sends data as frames.

Physical Layer: Transmits data as signals.

9.ANS:

Given:

Useful data = 80 MB = 80,000,000 bytes

Frame size = 1600 bytes

Overhead per frame = 80 bytes

1. Number of Frames

Answer: 50,000 frames

2. Total Overhead

Answer: 4 MB

3. Transmission Efficiency

Answer: 95.24%

4. Protocol Overhead

Protocol overhead adds extra bytes for communication.

More overhead reduces network efficiency.

Less overhead improves network performance.

10.ANS:

Given:

Original file = 240 MB

Compression = 30%

Segment size = 1400 bytes

Overhead = 60 bytes/frame

1. Compressed File Size

Answer: 168 MB

2. Number of Segments

Answer: 120,000 segments

3. Total Protocol Overhead

Answer: 7.2 MB

4. Final Data Transmitted

Answer: 175.2 MB

5. Multiple OSI Layers

Presentation Layer: Compresses data.

Transport Layer: Divides data into segments.

Data Link Layer: Adds overhead for reliable transmission.

Together, these layers ensure secure and reliable communication.